

Homework (Jalapeno)

- Q1.** The ancient Egyptians built the Great Pyramid with a slope of 3. What is the slope of a line perpendicular to it?
(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 3
(D) -3
(E) $\frac{2}{3}$
- Q2.** In ancient Greece, the philosopher Euclid used lines with slopes of 2 and $-\frac{1}{2}$. Are these lines perpendicular?
(A) Yes
(B) No
(C) Maybe
(D) Not enough information
(E) Only in some cases
- Q3.** The ancient Chinese built the Great Wall with watchtowers at a distance of 100 feet apart. If a line connecting two watchtowers has a slope of 4, what is the slope of a line perpendicular to it?
(A) $-\frac{1}{4}$
(B) $\frac{1}{4}$
(C) 4
(D) -4
(E) 2
- Q4.** What is the equation of a line perpendicular to $y = 2x - 1$ and passing through the point $(2, 3)$?
(A) $y = -\frac{1}{2}x + 4$
(B) $y = \frac{1}{2}x + 2$
(C) $y = 2x + 1$
(D) $y = -2x - 1$
(E) $y = x - 2$
- Q5.** The ancient Romans built roads with a slope of -3 . What is the slope of a line perpendicular to it?
(A) $\frac{1}{3}$
(B) $-\frac{1}{3}$
(C) 3
(D) -3
(E) 2
- Q6.** The ancient Mayans used a coordinate system with lines having slopes of 1 and 2. Are these lines perpendicular?
(A) Yes
(B) No
(C) Maybe
(D) Not enough information
(E) Only in some cases
- Q7.** What is the equation of a line perpendicular to $y = x + 2$ and passing through the point $(-2, 1)$?
(A) $y = -x$
(B) $y = x - 2$
(C) $y = 2x + 1$
(D) $y = -2x - 1$
(E) $y = -x + 1$
- Q8.** The ancient Babylonians used a number system with a base of 60. If a line has a slope of -2 in this system, what is its slope in the decimal system?
(A) $-\frac{1}{30}$
(B) -2
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$
(E) 2

Open Ended Questions (FRQ)

- Q9.** The ancient Egyptians built a temple with two perpendicular walls. One wall has a slope of -3 and the other has a slope of $\frac{1}{3}$. If the walls intersect at the point $(1, 2)$, find the equation of both lines.
- Q10.** The ancient Greeks built a theater with a circular stage. If a line from the stage to the audience has a slope of 2 and the radius of the stage is 10 feet, find the equation of a line perpendicular to it and passing through the center of the stage.

Chef's Tips

- Make sure students show their work and explain their reasoning
- Emphasize the importance of using the negative reciprocal to find the slope of a perpendicular line
- Have students use graph paper to visualize the lines and their relationships